Task 1: Guided Activity, Course registration

```
import Foundation


var subjects = ["Swift", "UIKit", "SwiftUI"]


subjects.append("Networking")

subjects.insert("Testing", at: 1)


if let index = subjects.firstIndex(of: "UIKit") {

   subjects.remove(at: index)

}


print("Sorted Subjects:", subjects.sorted())


var registered: Set<String> = ["Anu", "Ben", "Chitra"]


registered.insert("Anu")

registered.insert("Dev")


print("Registered Students:", registered)

print("Total Registered:", registered.count)


var marks: [String: Int] = [

   "Anu": 85,

   "Ben": 90,

   "Chitra": 78

]
```

```
marks["Dev"] = 88

marks["Anu"] = 95

print("Unknown Student Mark:", marks["Ravi"] as Any)

print("Student Marks:")

for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}

if let highest = marks.max(by: { $0.value < $1.value }) {
    print("Highest Mark: \(highest.value)")
    print("Top Student: \(highest.key)")
}
```

Output:

Output

```
Sorted Subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered Students: ["Ben", "Anu", "Dev", "Chitra"]
Total Registered: 4
Unknown Student Mark: nil
Student Marks:
Anu: 95
Ben: 90
Chitra: 78
Dev: 88
Highest Mark: 95
Top Student: Anu
```

Task 2: Practice Task, Word frequency counter

```
import Foundation

let sentence = "swift is fun and swift is fast"

let words = sentence.split(separator: " ").map { String($0) }


print("Total Words:", words.count)


let uniqueWords = Set(words)
print("Unique Words:", uniqueWords.sorted())


var frequency: [String: Int] = [:]

for word in words {
   frequency[word, default: 0] += 1
}

print("Word Frequencies:")

for word in frequency.keys.sorted() {
   print("\(word): \(frequency[word]!)")
}

let rankedWords = frequency.sorted {
   if $0.value == $1.value {
```

```
        return $0.key < $1.key
    }
    return $0.value > $1.value
}

print("Ranked Words:")

for item in rankedWords {
    print("\(item.key): \(item.value)")
}
```

Output:

```
Output

Total Words: 7
Unique Words: ["and", "fast", "fun", "is", "swift"]
Word Frequencies:
and: 1
fast: 1
fun: 1
is: 2
swift: 2
Ranked Words:
is: 2
swift: 2
and: 1
fast: 1
fun: 1
```

Task 3: Challenge Task, Set algebra and de-duplication

```
import Foundation

let swiftClub: Set<String> = ["Anu", "Ben", "Chitra", "Dev"]

let uiClub: Set<String> = ["Chitra", "Dev", "Esha"]

let bothClubs = swiftClub.intersection(uiClub)

print("Members in Both Clubs:", bothClubs.sorted())
```

```
let onlySwift = swiftClub.subtracting(uiClub)
print("Only Swift Club:", onlySwift.sorted())

let everyone = swiftClub.union(uiClub)
print("Everyone:", everyone.sorted())

let exactlyOne = swiftClub.symmetricDifference(uiClub)
print("Exactly One Club:", exactlyOne.sorted())

let numbers = [3, 1, 3, 2, 1, 4]
var seen: Set<Int> = []
var uniqueNumbers: [Int] = []

for number in numbers {
   if seen.insert(number).inserted {
      uniqueNumbers.append(number)
   }
}
print("Original Numbers:", numbers)
print("After Removing Duplicates:", uniqueNumbers)
```

output:

Output

```
Members in Both Clubs: ["Chitra", "Dev"]
Only Swift Club: ["Anu", "Ben"]
Everyone: ["Anu", "Ben", "Chitra", "Dev", "Esha"]
Exactly One Club: ["Anu", "Ben", "Esha"]
Original Numbers: [3, 1, 3, 2, 1, 4]
After Removing Duplicates: [3, 1, 2, 4]
```